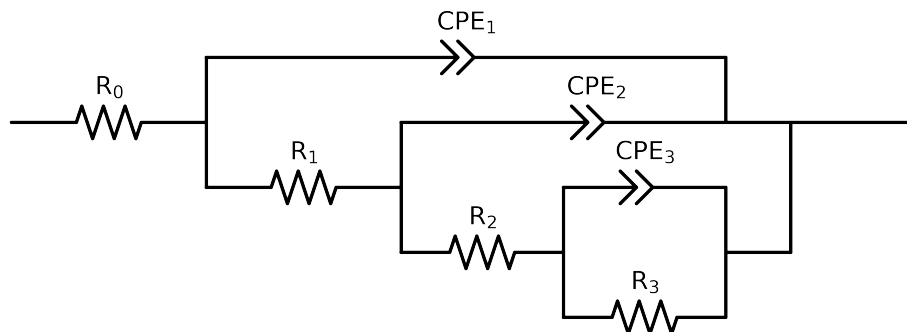


Comparative EIS Kinetics Report

Model Structure: R0-p(R1-p(R2-p(R3,CPE3),CPE2),CPE1)

Used data: original data



Element	Unit	Bounds	Cauvel_ A1_ 168H	Cauvel_ A1_ 1H	Cauvel_ A1_ 72H
R_0	Ω	$[1.0e+00, 1.0e+03]$	$9.61e+01 \pm 2.3e-01$	$8.29e+01 \pm 3.1e-01$	$1.08e+02 \pm 3.8e+00$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$1.46e+01 \pm 8.4e+00$	$4.25e+01 \pm 1.1e+01$	$8.63e+00 \pm 2.0e+00$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$3.84e+03 \pm 1.2e+02$	$1.43e+04 \pm 8.5e+02$	$2.28e+02 \pm 3.3e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$1.14e+04 \pm 3.3e+02$	$8.78e+03 \pm 1.1e+03$	$1.14e+04 \pm 4.3e+02$
Q_3	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$3.81e-04 \pm 1.2e-05$	$7.38e-04 \pm 1.7e-04$	$2.06e-05 \pm 3.6e-05$
n_3	-	$[5.0e-01, 1.0e+00]$	$6.72e-01 \pm 1.3e-02$	$1.00e+00 \pm 6.6e-02$	$8.63e-01 \pm 1.7e-01$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$8.01e-05 \pm 5.8e-05$	$1.31e-04 \pm 3.8e-05$	$5.39e-05 \pm 8.4e-05$
n_2	-	$[5.0e-01, 1.0e+00]$	$7.74e-01 \pm 5.5e-02$	$8.53e-01 \pm 3.1e-02$	$8.18e-01 \pm 1.6e-01$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$5.77e-05 \pm 5.9e-05$	$7.83e-05 \pm 4.0e-05$	$1.11e-04 \pm 5.3e-05$
n_1	-	$[5.0e-01, 1.0e+00]$	$7.80e-01 \pm 8.8e-02$	$8.32e-01 \pm 5.7e-02$	$5.47e-01 \pm 8.0e-02$
χ^2	-	-	$3.586e-05$	$1.898e-04$	$2.516e-04$

Analysis Results: Cauvel_A1_168H

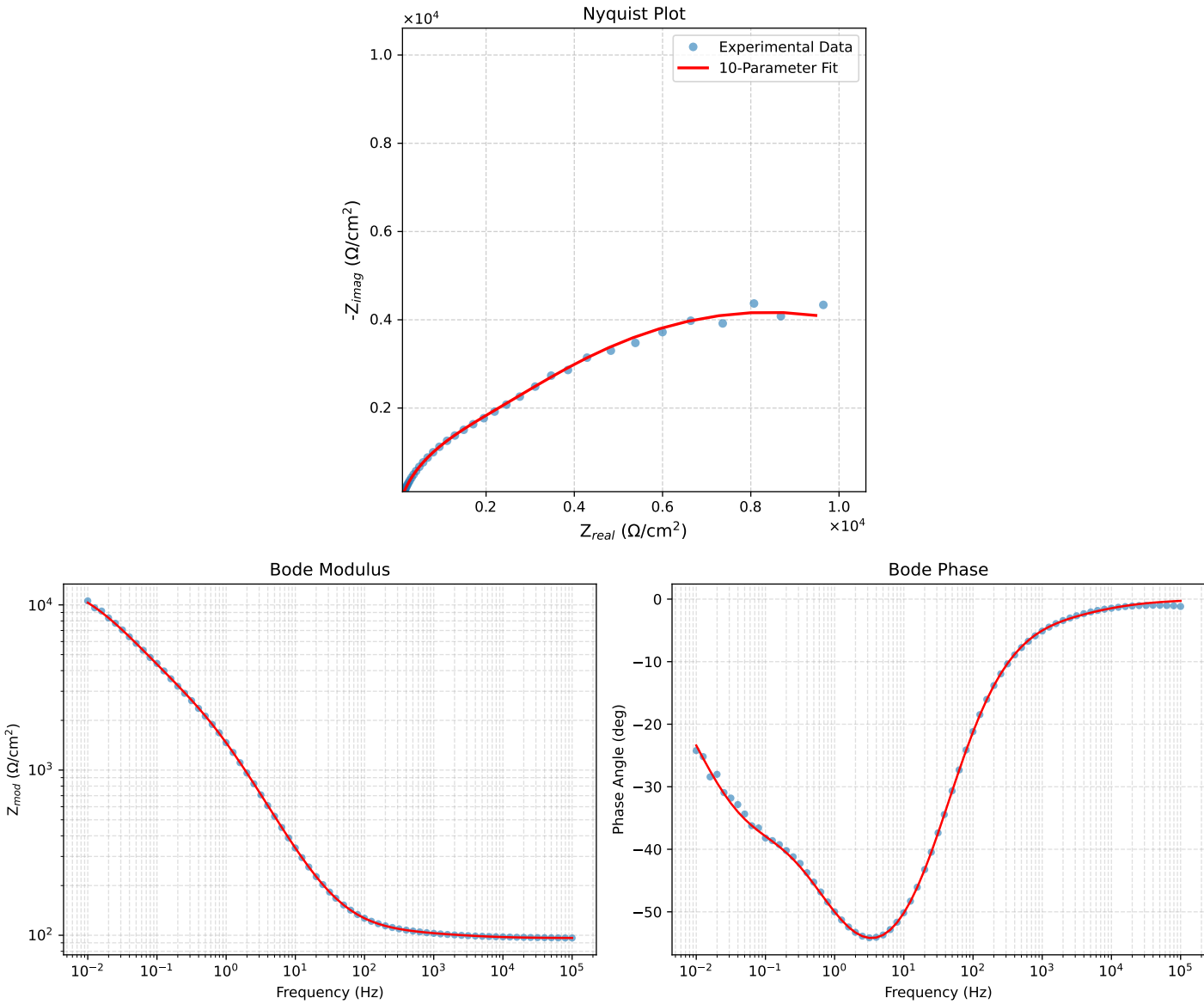


Figure 1: EIS Fit Results for Cauvel_A1_168H

Fitted Data: Cauvel_A1_168H

Frequency (Hz)	Z_{real} (Ω)	$-Z_{imag}$ (Ω)	Z_{mod} (Ω)	Z_{phase} ($^\circ$)
1.0002e+05	9.6319e+01	4.7959e-01	9.6320e+01	-0.2853
7.9512e+04	9.6359e+01	5.7048e-01	9.6361e+01	-0.3392
6.3105e+04	9.6409e+01	6.7866e-01	9.6411e+01	-0.4033
5.0215e+04	9.6469e+01	8.0463e-01	9.6472e+01	-0.4779
3.9902e+04	9.6543e+01	9.5331e-01	9.6548e+01	-0.5657
3.1699e+04	9.6634e+01	1.1273e+00	9.6641e+01	-0.6683
2.5137e+04	9.6747e+01	1.3312e+00	9.6757e+01	-0.7883
1.9980e+04	9.6885e+01	1.5645e+00	9.6897e+01	-0.9251
1.5879e+04	9.7053e+01	1.8319e+00	9.7070e+01	-1.0813
1.2598e+04	9.7259e+01	2.1379e+00	9.7282e+01	-1.2593
1.0020e+04	9.7504e+01	2.4784e+00	9.7536e+01	-1.4561
7.9282e+03	9.7804e+01	2.8661e+00	9.7846e+01	-1.6785
6.3079e+03	9.8148e+01	3.2837e+00	9.8203e+01	-1.9162
5.0347e+03	9.8538e+01	3.7341e+00	9.8609e+01	-2.1702
3.9931e+03	9.8991e+01	4.2399e+00	9.9081e+01	-2.4526
3.1441e+03	9.9509e+01	4.8142e+00	9.9625e+01	-2.7698
2.5195e+03	1.0003e+02	5.4059e+00	1.0018e+02	-3.0934
2.0008e+03	1.0061e+02	6.1026e+00	1.0079e+02	-3.4711
1.5807e+03	1.0124e+02	6.9283e+00	1.0148e+02	-3.9149
1.2536e+03	1.0190e+02	7.8887e+00	1.0220e+02	-4.4268
1.0007e+03	1.0259e+02	9.0043e+00	1.0298e+02	-5.0162
7.9003e+02	1.0337e+02	1.0417e+01	1.0389e+02	-5.7549
6.2881e+02	1.0420e+02	1.2073e+01	1.0490e+02	-6.6090
5.0223e+02	1.0512e+02	1.4043e+01	1.0605e+02	-7.6088
4.0015e+02	1.0618e+02	1.6444e+01	1.0745e+02	-8.8033
3.1550e+02	1.0747e+02	1.9484e+01	1.0922e+02	-10.2760
2.5202e+02	1.0889e+02	2.2948e+01	1.1129e+02	-11.9005
2.0032e+02	1.1061e+02	2.7194e+01	1.1390e+02	-13.8125
1.5801e+02	1.1272e+02	3.2472e+01	1.1730e+02	-16.0707
1.2556e+02	1.1517e+02	3.8612e+01	1.2147e+02	-18.5348
9.9734e+01	1.1810e+02	4.5967e+01	1.2673e+02	-21.2662
7.9449e+01	1.2159e+02	5.4631e+01	1.3330e+02	-24.1951
6.2919e+01	1.2589e+02	6.5233e+01	1.4179e+02	-27.3916
4.9867e+01	1.3109e+02	7.7846e+01	1.5246e+02	-30.7039
3.8422e+01	1.3824e+02	9.4885e+01	1.6767e+02	-34.4646
3.1250e+01	1.4514e+02	1.1096e+02	1.8269e+02	-37.3966
2.4934e+01	1.5421e+02	1.3157e+02	2.0271e+02	-40.4696
2.0032e+01	1.6486e+02	1.5507e+02	2.2634e+02	-43.2473
1.5625e+01	1.7971e+02	1.8666e+02	2.5911e+02	-46.0878
1.2467e+01	1.9634e+02	2.2061e+02	2.9533e+02	-48.3314
9.9734e+00	2.1643e+02	2.5979e+02	3.3814e+02	-50.2025
7.9719e+00	2.4111e+02	3.0552e+02	3.8920e+02	-51.7199
6.3516e+00	2.7183e+02	3.5923e+02	4.5048e+02	-52.8852
5.0134e+00	3.1130e+02	4.2382e+02	5.2587e+02	-53.7024
3.9860e+00	3.5850e+02	4.9558e+02	6.1166e+02	-54.1186
3.1758e+00	4.1576e+02	5.7606e+02	7.1043e+02	-54.1808
2.5161e+00	4.8738e+02	6.6843e+02	8.2724e+02	-53.9025
1.9955e+00	5.7387e+02	7.7010e+02	9.6041e+02	-53.3072
1.5879e+00	6.7602e+02	8.7915e+02	1.1090e+03	-52.4415
1.2581e+00	7.9918e+02	9.9811e+02	1.2786e+03	-51.3159
9.9777e-01	9.4211e+02	1.1229e+03	1.4657e+03	-50.0023
7.9274e-01	1.1044e+03	1.2513e+03	1.6890e+03	-48.5679
6.2903e-01	1.2879e+03	1.3840e+03	1.8906e+03	-47.0590
4.9888e-01	1.4912e+03	1.5200e+03	2.1294e+03	-45.5469
3.9860e-01	1.7056e+03	1.6551e+03	2.3766e+03	-44.1403
3.1672e-01	1.9423e+03	1.7988e+03	2.6474e+03	-42.8032
2.5148e-01	2.1977e+03	1.9509e+03	2.9386e+03	-41.5956
1.9998e-01	2.4705e+03	2.1126e+03	3.2506e+03	-40.5350
1.5879e-01	2.7675e+03	2.2888e+03	3.5914e+03	-39.5916
1.2601e-01	3.0926e+03	2.4808e+03	3.9647e+03	-38.7356
1.0016e-01	3.4481e+03	2.6868e+03	4.3713e+03	-37.9263
7.9503e-02	3.8450e+03	2.9077e+03	4.8207e+03	-37.0969
6.3140e-02	4.2867e+03	3.1372e+03	5.3120e+03	-36.1984
5.0123e-02	4.7802e+03	3.3692e+03	5.8482e+03	-35.1770
3.9833e-02	5.3256e+03	3.5924e+03	6.4239e+03	-34.0015
3.1638e-02	5.9270e+03	3.7964e+03	7.0386e+03	-32.6406
2.5126e-02	6.5799e+03	3.9675e+03	7.6835e+03	-31.0887
1.9957e-02	7.2754e+03	4.0924e+03	8.3474e+03	-29.3577
1.5849e-02	8.0014e+03	4.1601e+03	9.0183e+03	-27.4706
1.2590e-02	8.7402e+03	4.1630e+03	9.6810e+03	-25.4687
1.0001e-02	9.4743e+03	4.0988e+03	1.0323e+04	-23.3946

Analysis Results: Cauvel_A1_1H

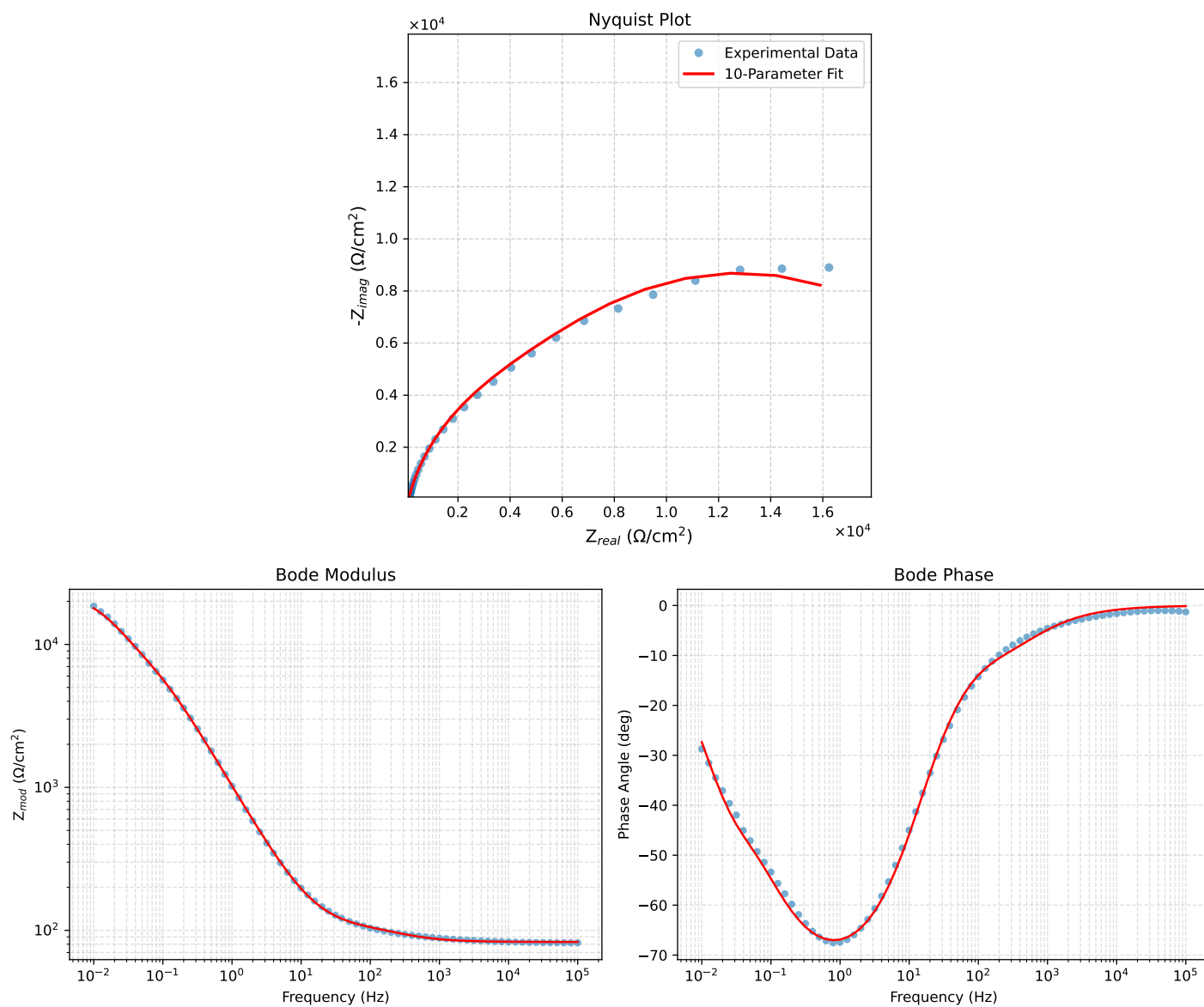


Figure 2: EIS Fit Results for Cauvel_A1_1H

Fitted Data: Cauvel_A1_1H

Frequency (Hz)	$z_{real}(\Omega)$	$-z_{imag}(\Omega)$	$z_{mod}(\Omega)$	$z_{phase}(\circ)$
1.0002e+05	8.2961e+01	1.8539e-01	8.2961e+01	-0.1280
7.9512e+04	8.2972e+01	2.2425e-01	8.2972e+01	-0.1549
6.3105e+04	8.2985e+01	2.7160e-01	8.2986e+01	-0.1875
5.0215e+04	8.3002e+01	3.2818e-01	8.3002e+01	-0.2265
3.9902e+04	8.3022e+01	3.9695e-01	8.3023e+01	-0.2739
3.1699e+04	8.3046e+01	4.8013e-01	8.3047e+01	-0.3313
2.5137e+04	8.3076e+01	5.8147e-01	8.3078e+01	-0.4010
1.9980e+04	8.3113e+01	7.0260e-01	8.3116e+01	-0.4843
1.5879e+04	8.3158e+01	8.4876e-01	8.3163e+01	-0.5848
1.2598e+04	8.3215e+01	1.0263e+00	8.3221e+01	-0.7066
1.0020e+04	8.3284e+01	1.2377e+00	8.3294e+01	-0.8514
7.9282e+03	8.3373e+01	1.4977e+00	8.3387e+01	-1.0291
6.3079e+03	8.3482e+01	1.8026e+00	8.3501e+01	-1.2370
5.0347e+03	8.3616e+01	2.1615e+00	8.3644e+01	-1.4808
3.9931e+03	8.3789e+01	2.6012e+00	8.3830e+01	-1.7781
3.1441e+03	8.4017e+01	3.1420e+00	8.4076e+01	-2.1417
2.5195e+03	8.4285e+01	3.7341e+00	8.4368e+01	-2.5367
2.0008e+03	8.4638e+01	4.4555e+00	8.4756e+01	-3.0134
1.5807e+03	8.5099e+01	5.3154e+00	8.5264e+01	-3.5742
1.2536e+03	8.5673e+01	6.2906e+00	8.5904e+01	-4.1994
1.0007e+03	8.6375e+01	7.3654e+00	8.6688e+01	-4.8740
7.9003e+02	8.7288e+01	8.6241e+00	8.7713e+01	-5.6425
6.2881e+02	8.8367e+01	9.9560e+00	8.8926e+01	-6.4282
5.0223e+02	8.9628e+01	1.1363e+01	9.0346e+01	-7.2253
4.0015e+02	9.1101e+01	1.2864e+01	9.2004e+01	-8.0373
3.1550e+02	9.2825e+01	1.4509e+01	9.3952e+01	-8.8837
2.5202e+02	9.4585e+01	1.6149e+01	9.5954e+01	-9.6888
2.0032e+02	9.6457e+01	1.7958e+01	9.8115e+01	-10.5463
1.5801e+02	9.8418e+01	2.0066e+01	1.0044e+02	-11.5240
1.2556e+02	1.0031e+02	2.2474e+01	1.0279e+02	-12.6286
9.9734e+01	1.0219e+02	2.5413e+01	1.0530e+02	-13.9650
7.9449e+01	1.0406e+02	2.9015e+01	1.0803e+02	-15.5797
6.2919e+01	1.0605e+02	3.3644e+01	1.1126e+02	-17.6017
4.9867e+01	1.0817e+02	3.9435e+01	1.1514e+02	-20.0300
3.8422e+01	1.1083e+02	4.7661e+01	1.2065e+02	-23.2684
3.1250e+01	1.1325e+02	5.5758e+01	1.2624e+02	-26.2121
2.4934e+01	1.1632e+02	6.6534e+01	1.3400e+02	-29.7690
2.0032e+01	1.1983e+02	7.9270e+01	1.4368e+02	-33.4843
1.5625e+01	1.2465e+02	9.7024e+01	1.5796e+02	-37.8953
1.2467e+01	1.3000e+02	1.1683e+02	1.7478e+02	-41.9453
9.9734e+00	1.3642e+02	1.4054e+02	1.9586e+02	-45.8529
7.9719e+00	1.4428e+02	1.6933e+02	2.2246e+02	-49.5658
6.3516e+00	1.5407e+02	2.0464e+02	2.5616e+02	-53.0254
5.0134e+00	1.6669e+02	2.4931e+02	2.9990e+02	-56.2333
3.9860e+00	1.8192e+02	3.0187e+02	3.5245e+02	-58.9257
3.1758e+00	2.0069e+02	3.6477e+02	4.1633e+02	-61.1818
2.5161e+00	2.2476e+02	4.4264e+02	4.9643e+02	-63.0795
1.9955e+00	2.5497e+02	5.3628e+02	5.9381e+02	-64.5719
1.5879e+00	2.9260e+02	6.4738e+02	7.1043e+02	-65.6782
1.2581e+00	3.4134e+02	7.8334e+02	8.5448e+02	-66.4547
9.9777e-01	4.0343e+02	9.4552e+02	1.0280e+03	-66.8935
7.9274e-01	4.8249e+02	1.1373e+03	1.2354e+03	-67.0111
6.2903e-01	5.8491e+02	1.3657e+03	1.4857e+03	-66.8160
4.9888e-01	7.1750e+02	1.6350e+03	1.7855e+03	-66.3061
3.9860e-01	8.8267e+02	1.9377e+03	2.1293e+03	-65.5098
3.1672e-01	1.0996e+03	2.2932e+03	2.5432e+03	-64.3829
2.5148e-01	1.3773e+03	2.6958e+03	3.0273e+03	-62.9383
1.9998e-01	1.7246e+03	3.1380e+03	3.5807e+03	-61.2071
1.5879e-01	2.1566e+03	3.6183e+03	4.2122e+03	-59.2034
1.2601e-01	2.6792e+03	4.1251e+03	4.9188e+03	-56.9974
1.0016e-01	3.2872e+03	4.6439e+03	5.6896e+03	-54.7077
7.9503e-02	3.9851e+03	5.1773e+03	6.5335e+03	-52.4136
6.3140e-02	4.7666e+03	5.7253e+03	7.4498e+03	-50.2207
5.0123e-02	5.6455e+03	6.2996e+03	8.4591e+03	-48.1342
3.9833e-02	6.6441e+03	6.8989e+03	9.5780e+03	-46.0780
3.1638e-02	7.8095e+03	7.5059e+03	1.0832e+04	-43.8644
2.5126e-02	9.1743e+03	8.0632e+03	1.2214e+04	-41.3120
1.9957e-02	1.0737e+04	8.4846e+03	1.3685e+04	-38.3160
1.5849e-02	1.2449e+04	8.6814e+03	1.5177e+04	-34.8897
1.2590e-02	1.4209e+04	8.5953e+03	1.6606e+04	-31.1709
1.0001e-02	1.5901e+04	8.2237e+03	1.7902e+04	-27.3471

Analysis Results: Cauvel_A1_72H

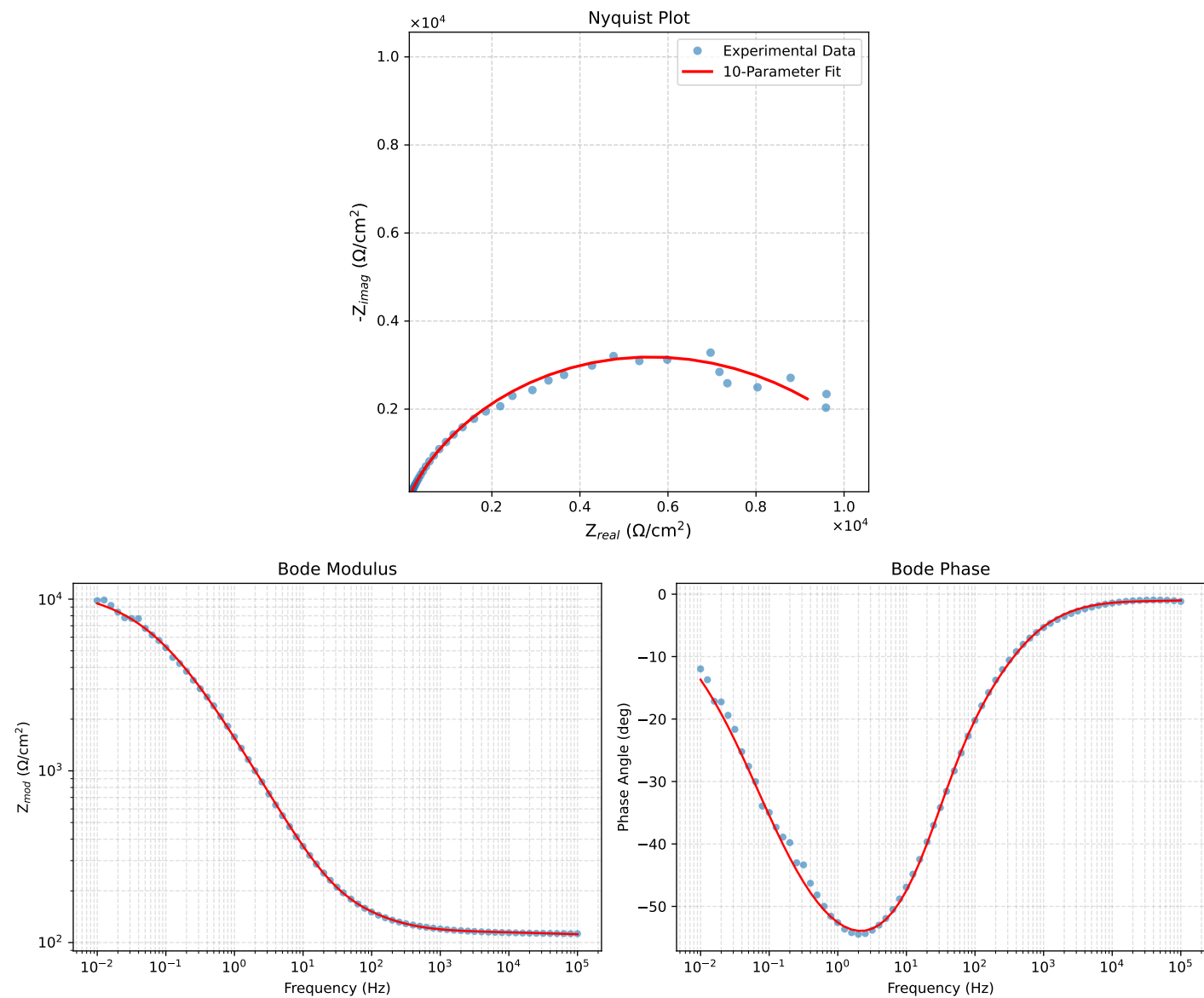


Figure 3: EIS Fit Results for Cauvel_A1.72H

Fitted Data: Cauvel_A1_72H

Frequency (Hz)	Z_{real} (Ω)	$-Z_{imag}$ (Ω)	Z_{mod} (Ω)	Z_{phase} ($^{\circ}$)
1.0002e+05	1.1174e+02	1.9534e+00	1.1176e+02	-1.0015
7.9512e+04	1.1204e+02	2.0196e+00	1.1206e+02	-1.0327
6.3105e+04	1.1235e+02	2.0781e+00	1.1237e+02	-1.0596
5.0215e+04	1.1266e+02	2.1296e+00	1.1268e+02	-1.0829
3.9902e+04	1.1297e+02	2.1784e+00	1.1299e+02	-1.1047
3.1699e+04	1.1327e+02	2.2290e+00	1.1329e+02	-1.1273
2.5137e+04	1.1357e+02	2.2879e+00	1.1359e+02	-1.1541
1.9980e+04	1.1385e+02	2.3617e+00	1.1388e+02	-1.1883
1.5879e+04	1.1413e+02	2.4599e+00	1.1416e+02	-1.2348
1.2598e+04	1.1440e+02	2.5938e+00	1.1443e+02	-1.2989
1.0020e+04	1.1466e+02	2.7721e+00	1.1469e+02	-1.3849
7.9282e+03	1.1493e+02	3.0148e+00	1.1497e+02	-1.5026
6.3079e+03	1.1519e+02	3.3249e+00	1.1524e+02	-1.6533
5.0347e+03	1.1546e+02	3.7166e+00	1.1552e+02	-1.8436
3.9931e+03	1.1577e+02	4.2259e+00	1.1584e+02	-2.0906
3.1441e+03	1.1612e+02	4.8855e+00	1.1622e+02	-2.4093
2.5195e+03	1.1648e+02	5.6397e+00	1.1662e+02	-2.7719
2.0008e+03	1.1693e+02	6.5957e+00	1.1712e+02	-3.2284
1.5807e+03	1.1748e+02	7.7819e+00	1.1774e+02	-3.7897
1.2536e+03	1.1813e+02	9.1864e+00	1.1849e+02	-4.4465
1.0007e+03	1.1890e+02	1.0811e+01	1.1940e+02	-5.1949
7.9003e+02	1.1989e+02	1.2826e+01	1.2058e+02	-6.1060
6.2881e+02	1.2106e+02	1.5112e+01	1.2200e+02	-7.1154
5.0223e+02	1.2245e+02	1.7728e+01	1.2373e+02	-8.2378
4.0015e+02	1.2414e+02	2.0781e+01	1.2587e+02	-9.5028
3.1550e+02	1.2626e+02	2.4460e+01	1.2861e+02	-10.9642
2.5202e+02	1.2861e+02	2.8451e+01	1.3172e+02	-12.4736
2.0032e+02	1.3140e+02	3.3115e+01	1.3550e+02	-14.1454
1.5801e+02	1.3468e+02	3.8672e+01	1.4012e+02	-16.0206
1.2556e+02	1.3827e+02	4.4925e+01	1.4539e+02	-17.9988
9.9734e+01	1.4230e+02	5.2261e+01	1.5159e+02	-20.1664
7.9449e+01	1.4675e+02	6.0827e+01	1.5886e+02	-22.5136
6.2919e+01	1.5192e+02	7.1312e+01	1.6782e+02	-25.1463
4.9867e+01	1.5784e+02	8.3863e+01	1.7873e+02	-27.9826
3.8422e+01	1.6569e+02	1.0097e+02	1.9403e+02	-31.3569
3.1250e+01	1.7309e+02	1.1722e+02	2.0905e+02	-34.1072
2.4934e+01	1.8269e+02	1.3819e+02	2.2907e+02	-37.1044
2.0032e+01	1.9386e+02	1.6219e+02	2.5276e+02	-39.9169
1.5625e+01	2.0930e+02	1.9451e+02	2.8573e+02	-42.9021
1.2467e+01	2.2647e+02	2.2928e+02	3.2227e+02	-45.3536
9.9734e+00	2.4700e+02	2.6943e+02	3.6551e+02	-47.4868
7.9719e+00	2.7193e+02	3.1632e+02	4.1714e+02	-49.3158
6.3516e+00	3.0254e+02	3.7154e+02	4.7914e+02	-50.8443
5.0134e+00	3.4130e+02	4.3828e+02	5.5549e+02	-52.0916
3.9860e+00	3.8693e+02	5.1308e+02	6.4263e+02	-52.9787
3.1758e+00	4.4158e+02	5.9812e+02	7.4347e+02	-53.5621
2.5161e+00	5.0925e+02	6.9768e+02	8.6376e+02	-53.8734
1.9955e+00	5.9063e+02	8.1034e+02	1.0027e+03	-53.9130
1.5879e+00	6.8713e+02	9.3554e+02	1.1608e+03	-53.7037
1.2581e+00	8.0513e+02	1.0782e+03	1.3457e+03	-53.2502
9.9777e-01	9.4575e+02	1.2356e+03	1.5560e+03	-52.5690
7.9274e-01	1.1118e+03	1.4066e+03	1.7930e+03	-51.6755
6.2903e-01	1.3095e+03	1.5923e+03	2.0616e+03	-50.5658
4.9888e-01	1.5424e+03	1.7900e+03	2.3629e+03	-49.2488
3.9860e-01	1.8048e+03	1.9894e+03	2.6861e+03	-47.7853
3.1672e-01	2.1143e+03	2.1973e+03	3.0493e+03	-46.1025
2.5148e-01	2.4684e+03	2.4037e+03	3.4454e+03	-44.2391
1.9988e-01	2.8640e+03	2.5997e+03	3.8679e+03	-42.2298
1.5879e-01	3.3050e+03	2.7799e+03	4.3187e+03	-40.0672
1.2601e-01	3.7867e+03	2.9353e+03	4.7911e+03	-37.7818
1.0016e-01	4.2977e+03	3.0570e+03	5.2741e+03	-35.4250
7.9503e-02	4.8365e+03	3.1405e+03	5.7667e+03	-32.9974
6.3140e-02	5.3884e+03	3.1807e+03	6.2571e+03	-30.5530
5.0123e-02	5.9444e+03	3.1762e+03	6.7397e+03	-28.1164
3.9833e-02	6.4498e+03	3.1284e+03	7.2045e+03	-25.7362
3.1638e-02	7.0182e+03	3.0408e+03	7.6486e+03	-23.4255
2.5126e-02	7.5202e+03	2.9190e+03	8.0668e+03	-21.2140
1.9957e-02	7.9887e+03	2.7701e+03	8.4553e+03	-19.1243
1.5849e-02	8.4202e+03	2.6013e+03	8.8128e+03	-17.1678
1.2590e-02	8.8120e+03	2.4201e+03	9.1383e+03	-15.3569
1.0001e-02	9.1647e+03	2.2328e+03	9.4327e+03	-13.6923